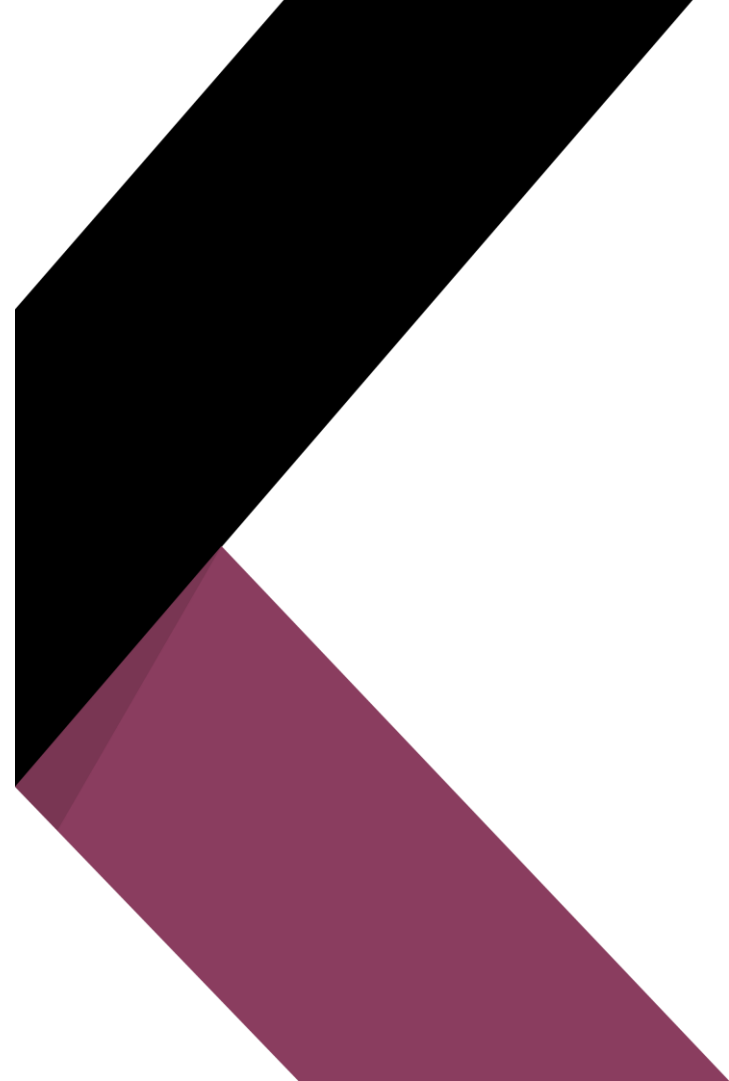


Procedure for drafting Test Case





Procedure for drafting Test Case

Test Case

Test

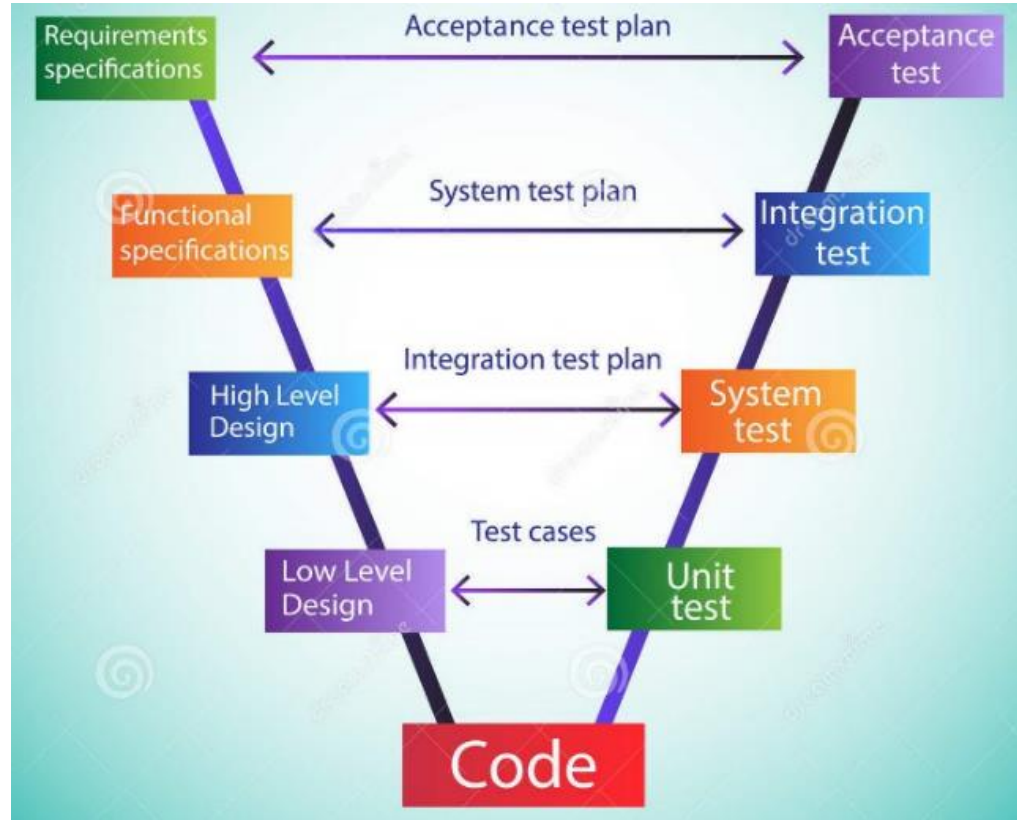
Case

The test cases are a set of conditions with which one goes to test if the requirements of a function are respected and if everything corresponds to what is required in the specifications. To be able to cover in a total way a requirement, for each test case must be contemplated as a 'positive test' that a 'negative test', that is, for each operating condition must be provided for the case in which the test covers a certain requirement and at the same time a test should be provided in which, if the opposite case is applied, this must give an opposite result to the previous case.

Only in this way are all possible cases covered and the test is robust.



Positioning in the V-Model





Test Case format -1-

A test is formalized in a document that describes in detail how it should be done. Each test can be set up like this:

- A unique identifier such as an alphanumeric number or string
- The name of the function to which it belongs
- The name of the project for which it is written
- The market to which it is addressed, for example if it is a market in Europe rather than Asia or America
- The type of test, for example if it is an integration or diagnostic test
- The preconditions, fundamental for the success of the test
- Actions, which describe what must be done starting from the preconditions described above
- The check to be carried out after the actions, ie the expected result of the test
- The result of the test, which can be OK or KO
- The name of the functional specification from which the test is extracted
- Any notes to complete the test in case the tester needs to add comments

Let's analyze a possible example of a test scheme below



Test Case format -2-

TEST ID	CATEGORY	PROJECT	MARKET	TEST TYPE	PRECONDITIONS	ACTIONS	EXPECTED RESULT	RESULT	NOTES	SPEC ref.	IMAGE
					This kind of test is valid for every Slaves nodes. Every node has a defined number. We describe a node with "X" where X can assume values from 0 to 31. This value is the node ID on the CAN Bus. There are 7 SLAVES: NPE, OBC, NFR --> wakeable and wakened BSM, HVB, OBC, P2, P4L, P4R, PCU --> unwakeable but wakened <u>CHECK THIS TEST FOR EVERY SLAVES</u>						
CCAN_TC022	Description Of MASTER States. S7-> S2 by E12	F173	All	System Integration	a) System supplied b) Key OFF c) failures are not present in the system and on CAN network	a) Perform a KeyON	a) C - STATUS_B_CAN.KeySts = ON (0x4) B - STATUS_NBC.KeySts = ON (0x4) The timer TkeyON (10s +/- 100ms) starts After TkeyON finish to count WAKE_C_NBC stops send messages WAKE_C_NBC.MainWakeUpSts = 1 All ignition counters are disable: TECMDetect, TSleepDelay, TMaxWake, TWaitBusSleep B - NVM_NBC.SystemCommand = 0 ("wU_wake_up_request")			F702_VH_R1	



Test Case format : integration test and diagnostic test -1-

We analyze two types of tests: integration tests and diagnostic tests.

An integration test is a test designed and written with a view to integration, ie a test whose result must respect what is provided by the software.

The following is an example of integration testing:

TEST ID	CATEGORY	PROJECT	MARKET	TEST TYPE	PRECONDITIONS	ACTIONS	EXPECTED RESULT	RESULT	NOTES	SPEC ref.
WindscreenIntelli gentRearWindscr eenWiperWasher	Interaction NBC and NBS to manage the telltale of the rear windscreen wiper state visualization			System Integration	a) System supplied b) Key OFF c) failures are not present in the system d) EOL Proxy : Start Byte 114, Start bit 0 used by NBC/NBS, (Rear windscreen wiper) checked by: Start Byte 114 Start bit 1 used by NBC, (Rear w/washer) ; checked by Start Byte 114, Start bit 5 used by NBC, (Rear windscreen wiper Speed Limit	a) Key OFF --> Key ON --> Crank --> Insert reverse gear Press the button on the steering wheel to activate the rear windscreen wiper and open the trunk	a) Verify that on dashboard doesn't appear the telltale of the rear windscreen wiper and NVO - BCAN - Rear/Window/WiperSwitchStz = 1 ("Active") The rear windscreen wiper is active during the period of reverse gear inserted	D		F147.V14_R2
WindscreenIntelli gentRearWindscr eenWiperWasher	Interaction NBC and NBS to manage the telltale of the rear windscreen wiper state visualization			System Integration	a) System supplied b) Key OFF c) failures are not present in the system d) EOL Proxy : Start Byte 114, Start bit 0 used by NBC/NBS, (Rear windscreen wiper) checked by: Start Byte 114 Start bit 1 used by NBC, (Rear w/washer) ; checked by Start Byte 114, Start bit 5 used by NBC, (Rear windscreen wiper Speed Limit	a) Key OFF --> Key ON --> Crank --> Insert reverse gear Don't press the button on the steering wheel to activate the rear windscreen wiper and open the trunk	a) Verify that on dashboard doesn't appear the telltale of the rear windscreen wiper and NVO - BCAN - Rear/Window/WiperSwitchStz = 0 ("NoActive") The rear windscreen wiper is active during the period of reverse gear inserted	D		F147.V14_R2



Test Case

format : integration test and diagnostic test -2-

A diagnostic test is a test designed and written with a view to diagnosis, ie a test whose result must respect what is provided for by the CDD.

The following is an example of a diagnostic test:

TEST ID	CATEGORY	PROJECT	MARKET	TEST TYPE	PRECONDITIONS	ACTIONS	EXPECTED RESULT	RESULT	NOTES	SPEC ref.	IMAGE
Lighting.TC0023	Front/Rear ceiling lights and spotlights management.Logic working description	F1T3	All	Diagnostic	System is supplied There aren't failure in the system Vehicle is in Key OFF Internal lights are on OFF	a) Press the button of every light, left and right front lights and left and right rear lights, one by one, for more than 60s	a) verify: Lights turn off and NBC stores a DTC			F078_Vf19_R 3 F083_Vf18_R 1	
Lighting.TC0024	Front/Rear ceiling lights and spotlights management.Logic working description	F1T3	All	Diagnostic	System is supplied There aren't failure in the system Vehicle is in Key OFF Internal lights are on OFF	a) Simulate an undervoltage on the LCC: LIN - CeilingToNBC.ResponseErrorCeiling_Overtemp = 1 (Error Present) and push one button to turn on one light in the car	a) Light turn on and NBC stores a DTC			F078_Vf19_R 3 F083_Vf18_R 1	



From the functional requirement to the test case -1-

To get to the writing of a test case we start from the functional requirements, described in the specification or VF.

It starts from the analysis of the specification which is one of the most important steps useful for the clean writing of the tests:

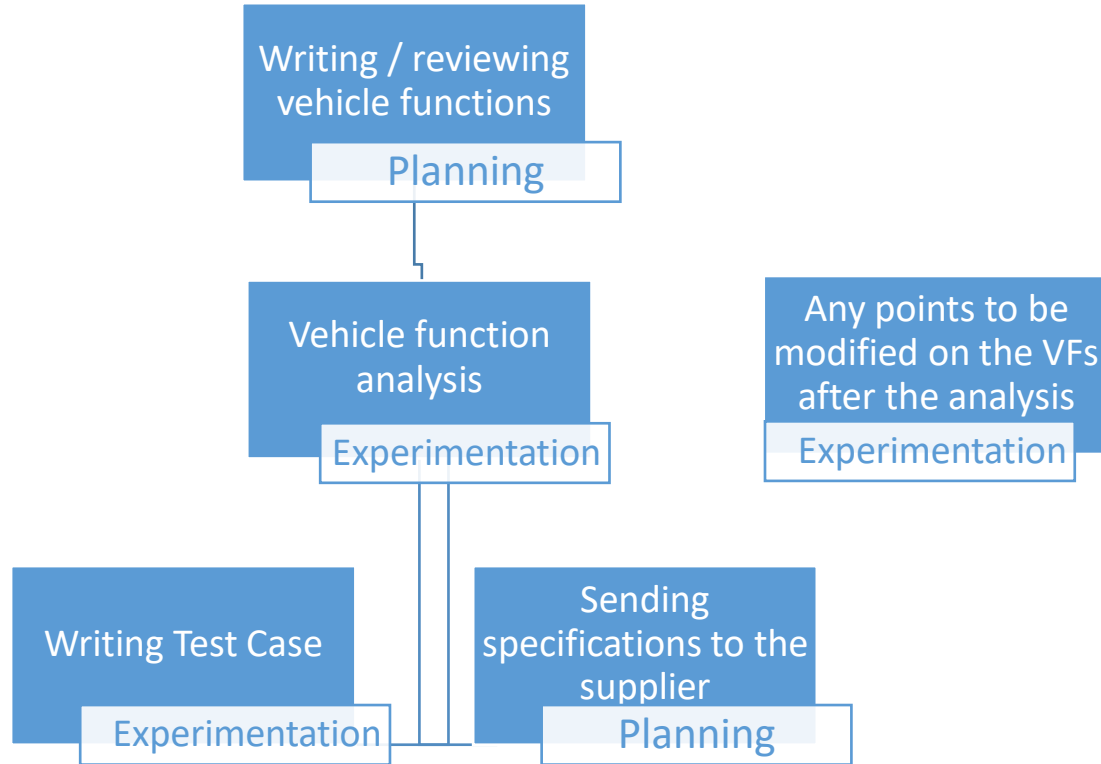
for VF analysis, we mean the study process of a vehicle function and any changes to be made to the test, following an accurate comparison between the requirements themselves, DB Files, already performed test results, related documentation.

The changes can be of various kinds, from the wrong nomenclature of a signal with respect to the DB File where it is contained (or even to its absence), parts of function developed on exchange of internal messages that can not be tested with a view to integration testing.

Before you can get to write the test there is a procedure to be followed so that the test cases are correct and functional, shown below.



From the functional requirement to the test case -2-





From the functional requirement to the test case -3-

One must assume that the tests have to 'cover' a totally specific, that is, you must write all tests useful to test all that is written in a VF and predict, as we said before, even against all cases.

It can happen that downstream of the analysis of the requirements you realize that there may be additional test cases and that, if plausible, must still be written, even if not explicitly present in VF.

There are also cases in which there are requirements in VF, with which are described the 'internal' behavior to the component that can not be monitored. That type of test, identified as a 'component test', is the responsibility of the supplier of the component itself who is concerned with carrying out its tests.



From the functional requirement to the test case -4-

Let's see in detail how, starting from a requirement, we get to writing the test.

The requirement can be a description of the operation presented in the form of text, table, state machine, logical equation, etc.

We analyze the case of a requirement extracted from a VF whose object is the CCAN WakeUP Management.

Let's take as an example a state machine that describes its behavior.



CAN-C Ignition-Off Bus
Wake Up, Active, and Sleep
Feature: **MASTER ECU**





The diagram shows the logic for the Ignition On state. A yellow box labeled 'S2: Ignition On' has an arrow pointing to a circle labeled 'S7: WaitBus-Sleep'. The circle contains the text 'ns/Start' and 'TwaitBusSleep timer; Stop message transmission'. An arrow labeled 'Clear' points to a small square box in the circle. Another arrow points from the circle to a small square box labeled 'S2: Ignition On'.

[illegible]

kinetom





From the functional requirement to the test case -8-

Let's illustrate another example to get to write test cases starting from a description of the operational expectation.

4.1.4.2 Wake up Initiated by the Master

If the Master is awakened by a Hardware Wake up that requires the CAN C bus to be awake, it shall transmit the WAKE_C_NBC message, setting the MainWakeSts bit and the NodeX bit corresponding to itself. Similarly, the Master shall also send its WAKE_C message with active MainWakeSts and NodeX bits, when it recognizes that the CAN BH bus is active. Figure 2 below shows a skeleton diagram of the Class B Master State Machine as defined in TFO 07320, Figure 2. The diagram below is a reference to show the ADDITIONAL activities that the Master must perform to coordinate waking the CAN C busses when the CAN BH bus is awake. (Note: putting both the CAN C busses and the CAN BH bus to sleep will be addressed in state chart describing the CAN C Master later in this document.)

<Picture>

Finally, if the Master has received a WAKE_C_<ECU> from an ECU, it shall immediately transmit its own WAKE_C_NBC message. If that slave's message contained an active WakeReq signal, then the Master shall set that slave's NodeX bit in its NodeAwake "array." It shall also set the MainWakeSts bit. The Master shall populate the WakeRsn signal with the **specific** reason that the master is awake. It shall also service the WakeCntr signal, incrementing it with each the Master is awakened (up to 255 per ignition-off period.) The master shall wakeup and send out the master wake message in sufficient time to support the necessary ignition-off features. See the vehicle specific CAN C NOS document [1] for details. The Master shall not go to sleep sooner than <T_{EMCDetect}> upon an invalid wakeup (i.e., an EMI pulse).

TEST ID	CATEGORY	PROJECT	MARKET	TEST TYPE	PRECONDITIONS	ACTIONS	EXPECTED RESULT	RESULT	NOTES	SPEC ref.	IMAGE
					This kind of test is valid for every Slaves nodes. Every node has a defined number. We describe a node with "X" where X can assume values from 0 to 31. This value is the node ID on the CAN Bus. There are 7 SLAVES: NPE, OBC, NFR --> wakeable and wakened BSM, HYB, OBC, P2, P4L, P4R, PCU --> unwakeable but wakened CHECK THIS TEST FOR EVERY SLAVES						
PowerManagement.CAN-C WakeUpStrategy. TC006	FunctionalRequirements.WakeUpInitiatedByTheMaster	FT03	All	System Integration	a) System supplied b) Key OFF c) failures are not present in the system and on CAN network	a) Perform a CC on CCAN bus in order to have a noise (distrub) on the network that cause a false wake on the Body	a) Verify : 1) WAKE_C_NBC.WakeRsn = the number associated to the reason from 0x00 to 0x7F WAKE_C_NBC.MainWakeSts = 0 WAKE_C_NBC.WakeReq = 0 WAKE_C_NBC.WakeCntr starts to count Within T _{ECN} the Body go to sleep			FT02_V1_R1	



From the functional requirement to the test case -9-

Let's illustrate another example to get to write test cases starting from a description of the operational expectation.

4.1.6 End of Normal Operation / Transition to Sleep

4.1.6.1 Master

Ten-seconds after the master sees that no ECU on any bus (including the NBC itself) requires the bus to be awake, it shall set its Master wake status flag inactive, resend the master wake status message, and start the sleep timer, <TWaitBusSleep> (i.e., enter the TWaitBusSleep State).

In the case of the Ignition having been turned off, the Master shall fulfill the 10-second and 10-minute timer requirements of TFO 07321 from State S6 (Ignition Off Awake Mode).

During the time <TWaitBusSleep>, only one of the following conditions shall cause the NBC to transition out of the sleep process:

- 1) WAKE_C<ECU>.WakeReq. or
- 2) The CAN BH bus becomes active
- 3) An active Hardware Wake-up input on the NBC.

While in State 6, Ignition Off Awake Mode, the master shall maintain the timer, T_{SleepDelay}, to enforce a 10-second time since the CAN C bus is needed to be awake. All of the following must be true for 10 seconds:

- 1) no wakeable slaves have an active WakeReq bit in their WAKE_C<ECU> message,
- 2) no NON-wakeable slaves have sent a STATUS_C<ECU> message (present just after the ignition is turned off),
- 3) no CAN BH slaves have their Active Load (AL) bits set, AND
- 4) no active Hardware wakeup inputs on the NBC

Note: any slave that fails to return its WAKE_C<ECU> message within 20 mS of receiving the WAKE_C_NBC message, the Master shall clear that slave's NodeX signal (where x corresponds to a Slave's node ID).

The NBC shall force the CAN C bus to transition to the WaitBusSleep State if 10 minutes have elapsed since State S6 (Ignition Off Awake Mode) was entered. At that time it shall command the CAN BH bus to go to NET_OFF state.

TEST ID	CATEGORY	PROJECT	MARKET	TEST TYPE	PRECONDITIONS	ACTIONS	EXPECTED RESULT	RESULT	NOTES	SPEC ref.	IMAGE
					This kind of test is valid for every Slaves nodes. Every node has a defined number. We describe a node with "X" where X can assume values from 0 to 31. This value is the node ID on the CAN Bus. There are 7 SLAVES: NPE, OBC, NFR -> wakeable and wakened BSM, HYB, OBC, P2, P4L, P4R, PCU -> unwakeable but wakened CHECK THIS TEST FOR EVERY SLAVES						
PowerManagementCAN-CWakeUpStrategyTC007	EndOfNormalOperationTransitionToSleepMaster	F113	All	System Integration	a) System supplied b) Key OFF c) Failures are not present in the system and on CAN network d) Master in NOT in sleep state	a) No Slave's ECU sends any messages on the network for 10s (the case when, for each slaves, 'WAKE_C_'<ECU>.'WakeReq = 0)	a) verify that the Body sets WAKE_C_NBC.MainWakeSts = 0 and within this time T _{WaitBusSleep} the Body goes to sleep			F702_Vr1R1	



Test procedure -1-

When all the tests for a VF are written, proceed with the actual tests.

The test phase can be performed in several ways:

1. In the car
2. At the bench
3. At the simulator

This depends on the type of test to be performed.



Test procedure -2-

1. It is preferred to do a test in the car when it is necessary to see, on the user's side, the real behavior of a given action. For example, if you want to test the behavior of the windscreen wipers at a high vehicle speed, so as to verify the user's real behavior in a given condition.
2. We proceed with the bench tests when it is necessary to simulate / measure the electric / electronic behavior, for example when you have to carry out tests involving currents and / or voltages, also by means of an oscilloscope
3. We test the simulator when you do not have a car or a bench, or when you need the simulation of CAN messages, whose dynamics are very fast.

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